

a Allocation problem

One native record, matched conditions

Supply, reserve, efficiency and capacity
Same signed energy and cyclic inventory



Coarse model

Interval mean
Exact optimum



Compare against no reserve, H_0

Coarse gain + native upper below H_0
→ aggregation false positive



Native model

Every second
Feasible + upper



Finite-capacity feasibility uses native chronology.

b Conservative screen

1 Summarize each decision interval

Sort native values; retain K rank means



2 Evaluate an optimistic upper

Nonnegative dual multiplier → U_K^+



3 Apply the screening rule

$U_K^+ < H_0$: exclude a gain
Otherwise: inconclusive

Optional: endpoints → compression enclosure.

Dispatch or positive gain: native replay required.